Wael Bahgat

Principal Software Engineer

Bethesda, MD | wael.bahgat@gmail.com
wbahgat.github.io/resume | linkedin.com/in/wael-bahgat-5667662

SUMMARY

Principal software engineer with 15+ years of C++ application and server-side software for satellite communications, aviation surveillance, maritime autonomy and telecom. Designs frameworks with a stable core that other engineers extend without changing shipped code; puts in place the simulation and release pipelines that let teams test every day and ship what was tested; leads delivery from customer requirements to production.

HIGHLIGHTS

- Designed and built from the first line a surveillance system framework with a stable core that other engineers extend without changing it; the approach was adopted across an aviation surveillance department. Has since applied it to maritime autonomy and a satellite core network.
- Twice replaced slow, costly testing on real systems with simulation: a playback simulator for an autonomous vessel, and software-in-the-loop simulation for satellite ground software.
- Led a team of five engineers through a full delivery cycle, from customer requirements to deployment; mentors engineers and takes part in hiring.

CORE SKILLS

Languages: C++, Qt, Java, SQL and PL/SQL, Dart/Flutter; Python for scripts and tooling
Architecture: Object-oriented design, plugin frameworks, SOLID and the open/closed principle, design patterns, multithreading, thread and session pools, REST and TCP services, interface definitions between systems
Testing: GTest, GCov, test-driven development, software-in-the-loop and playback simulation, performance and load testing
Delivery: Docker, GitLab CI/CD, release engineering, amd64/arm64 cross-compilation, Linux, systemd, gdb
Data and cloud: PostgreSQL, MongoDB, Oracle, SQLite; AWS (ECS Fargate, RDS, S3, CloudFront)

EXPERIENCE

Principal Software Engineer, Lead Developer, Core Network

Dec 2023 – Present

Elveo Mobile (formerly Lynk Global) · *Direct-to-device satellite operator. Joined as Senior Network Software Engineer; promoted to Principal in 2026.*

- Co-designed and built the plugin architecture of the dual-mode (satellite and ground) GSM/LTE core network software in C++: features are separately loaded modules around a stable core, so capability is added without changing shipped code. Have led its development since.
- Designed the framework's core libraries: per-module configuration, asynchronous thread-pool processing, a generic database session pool, and interchangeable back ends (PostgreSQL, MongoDB, SQLite).
- Created the container build system and a development environment produced from the same definitions as production, for amd64 and arm64; adopted as the team's standard way of developing.
- Designed and own the GitLab CI/CD release pipeline: unit tests and coverage, multi-stage Docker builds, release candidates, and promotion of the tested candidate to release.
- Built internal operator tooling (web front end, REST back end, PostgreSQL, command-line tooling) and introduced software-in-the-loop simulation for testing; mentored an intern through its implementation to delivery.
- Integrated GTest and GCov across all components; take part in the team's hiring interviews.

Senior Software Specialist

Sep 2021 – Dec 2023

L3Harris · *Aviation surveillance.*

- Built core C++ components for aviation surveillance systems.

- Designed a framework for multiple service applications sharing a common core with extendable layers: configuration manager, log manager, thread pool, pluggable business logic and XML-driven configuration.
- Applied test-driven development with GTest, coverage analysis with GCov, and performance testing with custom profilers.

Development Team Lead
Jul 2020 – Sep 2021

Gibbs & Cox, a Leidos company · *Proof-of-concept phase of an autonomous navigation system for surface vessels.*

- Learned the maritime domain from the existing code and from the vessel itself, and led the team of five software engineers within a year: customer requirements, architecture, implementation, testing and deployment.
- Designed a modular Qt/C++ framework that models onboard subsystems as generic components behind a hardware abstraction, so the software is not tied to one vessel.
- Built a playback simulator from recorded data, so use cases run at a desk instead of at sea, and an operator interface for development and testing.

Senior Software Specialist
May 2016 – Jul 2020

L3Harris · *Aviation surveillance.*

- Aggregated radar and ADS-B data in real time across multiple sites, and developed Qt/C++ air traffic display applications.
- Designed and built a new surveillance system from scratch on a framework of decoupled components with a thread pool, central logging and configuration-driven behavior; the approach was later adopted across the department's projects.
- Built a drone surveillance user interface in C++/Qt, and reusable C++ libraries with GTest coverage.
- Member of the team that developed a voice switching solution for air traffic control, supporting hybrid analog and VoIP calls.

Lead Software Engineer, Production Engineering
Jul 2012 – Apr 2016

Openet Telecom

- Delivered hotfixes and patches across the production stack in C++ and Java for carriers including Verizon, AT&T and Sprint.
- Diagnosed memory leaks, deadlocks and crashes with gdb and mdb, and took critical customer issues from first report to delivered patch.

Software Engineer to Lead Software Engineer, Professional Services
Jan 2005 – Jun 2012

Openet Telecom

- Designed and delivered telecom mediation and billing solutions for major carriers in Java and PL/SQL on high-volume Oracle databases.
- Authored requirements, performance test specifications and architecture documents.

INDEPENDENT PROJECT

Yalabeena, a marketplace for booking sessions with hosts (yalabeena.com)

- Designed, built and operate the whole product alone: one Flutter code base for web, Android, iOS and desktop in English and Arabic, a C++ REST back end built from plugin modules, and PostgreSQL. Includes payments, in-app video sessions and chat, push notifications and an assistant.
- More than 200 end-to-end test files across four platforms, and a release-candidate process that proves each release on the web before it reaches the app stores.

EDUCATION

M.S. Electrical Engineering · NYU Tandon School of Engineering, Brooklyn, NY — GPA 3.87

B.S. Computer Engineering · Ain Shams University, Cairo, Egypt